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A backstepping disturbance observer control for multirotor UAVs: theory and experiment

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ABSTRACT
This paper presents a backstepping disturbance observer based control (DOBC) for trajectory tracking motion control of multirotor Unmanned Aerial Vehicles (UAVs). Two disturbance observers (DO) estimate external force and torque disturbances acting on the UAV. A nonlinear backstepping dynamic state feedback is proposed which uses the DO estimates to achieve exponential stability for the closed-loop tracking error under constant disturbances. The stability analysis accounts for the entire nonlinear vehicle model which includes translational and rotational dynamics. Software-in-the-loop simulation and experimental results are presented showing the effectiveness of the proposed method using the commonly used PX4/Pixhawk development framework.

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1. Introduction

Multirotor Unmanned Aerial Vehicles (UAVs) are popular due to their simple robust electro-mechanical design, high manoeuvrability, and low-cost. They are used in environment monitoring, terrain mapping, emergency response, and other applications (Kendoul, 2012). Increased adoption depends on a high-performance motion controller which can track desired 3D position and yaw trajectories. There are a number of challenges involved in improving motion control given the system's underactuated nonlinear dynamics, bounded input, unmeasured and time-delayed states, and external disturbances. These challenges have attracted significant interest from the research community. This paper focuses on improving the robustness of nonlinear motion control to external force and torque disturbances. This robustness is clearly important for improving the performance and safety in a range of environments. For example, in outdoor applications, the UAV is subject to disturbances such as wind-gusts or changes in thrust due to nearby obstacles. As confirmed by the interest shown in recent literature on the topic, improving the robustness of motion control is clearly an important practical and theoretical problem.

Many linear and nonlinear motion control laws have been proposed (Hua et al., 2013). Typically linear control is based on an approximate linear model. Examples of linear designs include Proportional-Integral-Derivative (PID) control in Hua et al. (2013) or Linear Quadratic (LQ) optimal control in Liu et al. (2016) and Nonami et al. (2010). Since linear designs lead to a local convergence result, nonlinear control methods, which directly account for system nonlinearity, have been proposed to achieve a wider operating range. Examples of nonlinear methods include backstepping in Castillo et al. (2005), feedback linearisation in Koo and Sastry (1998), sliding mode in Reinoso et al. (2016), and differential flatness in Kumar (2011), Faessler et al. (2018) and Poultney et al. (2018). However, most of the proposed methods suffer from a lack of any rigorous analysis of robustness to external disturbances. Some exceptions which address robustness include adaptive techniques (Cabecinhas et al., 2014; Goodarzi et al., 2015), integral state augmentation (Bouabdallah, 2007; Omari et al., 2013), sliding surfaces (Madani & Benallegue, 2006; Sumantri et al., 2013), or disturbance observer-based control (DOBC) (Dong et al., 2014; He et al., 2014; Lee et al., 2016; Shao et al., 2018). We review the relevant subset of these papers below which are related to the proposed nonlinear DOBC.

We remark that many proposed methods have an inner-outer loop control structure which separates the design for the translational and rotational subsystems. This leads to a simple design and is found in field-tested autopilots (Meier, 2018). However, most methods which have this structure do not analyse the effect of inner-outer loop coupling on the stability of the *entire* (i.e. combined rotational and translational) system. The method we propose does not take an inner-outer loop structure and features a stability analysis which accounts for the entire dynamics.

A number of researchers have applied adaptive control to improve robustness. An adaptive backstepping controller is developed in Cabecinhas et al. (2014) with asymptotically stable tracking error for constant force disturbances. The state of the system is augmented with thrust and its time derivative, and overparameterization of the disturbance estimate leads to two third-order update laws. A saturation function limits thrust and a projection operator avoids wind-up of force estimates. The overparameterization, projection operator, and saturation significantly increase the complexity of the control. The method is validated experimentally with the controller implemented on a ground computer using Matlab. Work in Goodarzi et al. (2015)

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designs an adaptive nonlinear control which compensates for disturbances. To avoid the singularities of Euler angles or the unwinding phenomenon of quaternions, the controller is developed on SE(3), i.e. it uses rotation matrices to represent attitude. Separate attitude and position control modes are proposed. In both modes, adaptive laws compensate the effect of constant disturbances. The position mode provides position tracking by relating the direction of desired thrust to the position error. This direction defines a column of the desired rotation matrix. Almost global attractiveness is guaranteed in the case of constant disturbances provided certain gain conditions are satisfied. The authors do not consider the effect of mode switching on the stability of the hybrid system. The method is validated in simulation and experiment but the effect of external disturbances is not investigated.

Integral augmentation is another method used for improving robustness and is combined with nonlinear control with an inner-outer loop structure in Omari et al. (2013). The effect of rotor drag and blade flapping is included in the dynamic model and controller design. The outer loop controller is designed to control position and linear velocity. To achieve robustness against model uncertainty and external disturbances, saturated integrators with fast desaturation are employed. The approach is experimentally validated for various reference trajectories and in the presence of wind forces. However, only asymptotic (as opposed to exponential) stability is proven for the outer loop design and the effect of inner-loop tracking error on closed-loop performance is not considered.

Sliding mode control is commonly used to improve robustness in a range of applications. A nonlinear sliding surface (NSS) method with inner-outer loop structure is presented in Sumantri et al. (2013). Experimental results are presented which include an external disturbance from a small fan. Motion is constrained as the UAV is attached to a stand for safety. Work in Madani and Benallegue (2006) combines backstepping and sliding mode techniques. The dynamics is divided into four subsystems: lateral position, altitude, yaw, and actuator dynamics. To avoid complex expressions for the control law, low-pass filtered derivatives are employed to estimate derivatives of the virtual control. Asymptotic stability of the closed-loop is derived without accounting for subsystem interaction. The sign function used in the sliding surface equation is approximated by a continuous function to address chattering. However, the effect of this approximation is not investigated in the stability proof. The method is validated on a stand which limits motion to vertical and yaw DoF.

DOBC is presented in a number of papers. Work in He et al. (2014) combines a DO with backstepping for a traditional helicopter UAV. Low-pass filtered derivatives of virtual controls are employed for angular acceleration estimation. Filtering estimation error leads to ultimate boundedness of tracking error. Work in Shao et al. (2018) integrates an inner-outer loop backstepping control with an extended state observer (ESO) to account for parametric uncertainty and disturbances. Linear and angular velocity are assumed unmeasured and estimated by the ESO alongside the disturbance. For simplifying the control expressions, dynamic surface control is used. The inner-loop control relies on simplified rotational dynamics. The design yields ultimately bounded tracking error for time-varying disturbances. The method is validated in simulation. Work in Dong et al. (2014) proposes a DOBC with backstepping and inner-outer loop structure. Simplified rotational dynamics is considered and the translational system is approximated near hover. A range of system uncertainty is considered: unstructured model error, input delay, and external disturbances. Ultimate boundedness is proven and simulation and experimental results are presented using a fan as a disturbance. Work in Lee et al. (2016) presents a linear DOBC for hover control. The inverse of the model's transfer function is used to estimate the external force disturbance. This estimate is used in an inner-outer loop control structure. Stability analysis is performed using the Small Gain Theorem. Simulation and experimental results are presented including a comparison with backstepping without a DO.

In this work, we propose a nonlinear backstepping DOBC for multirotor UAV trajectory tracking. The DO improves the closed-loop's robustness to external force and torque disturbances. Although the idea of using a DO and backstepping state feedback is not new e.g. Cabecinhas et al. (2014) and Shao et al. (2018), the particular closed-loop structure of the proposed DOBC is a main contribution. Specifically, the DO error dynamics is LTI, globally exponentially stable, and depends only on disturbance estimate error. The disturbance estimate error feeds the tracking error dynamics in a cascade interconnection. This cascade structure makes the stability analysis, implementation, and controller tuning straightforward. In particular, the convergence of the tracking error is proven by analysing the system dynamics in the error coordinates defined from the backstepping procedure. It is shown that the tracking error dynamics is Linear Time-varying (LTV) with the disturbance estimate error acting as a system input. Using LTV system stability analysis, *exponential* stability is proven in the presence of constant disturbances. Exponential convergence is an important property since it ensures robust closed-loop stability and ultimate boundedness of the tracking error in the case of time-varying disturbances. The simplicity of the proposed control is seen in comparison with our previous work (Moeini et al., 2019) where a more general controller structure is used. Here, expressions for the control are too complex for on-board implementation and only Matlab simulation results are presented.

Another important aspect of this work is that the control design relies on the entire system dynamics, i.e. the coupled translational and rotational rigid body dynamics. This avoids having to make common simplifying assumptions typical of designs with inner-outer loop structure (e.g. linear approximation of the rotational dynamics) during the closed-loop stability analysis of the entire rotational and translational system dynamics (Martin & Salaün, 2010). Although designs based on the entire dynamics exist in the literature, such designs do not explicitly address robustness to disturbances (Poultney et al., 2019) or provide experimental testing (Cabecinhas et al., 2014).

Another important contribution of this work is the validation performed in simulation and experiment using the standard PX4 autopilot under realistic flight conditions. The PX4 firmware is a widely-used and open source project intended for research and commercial use. It runs on a number of hardware platforms including the Pixhawk family. Although the

proposed design was simulated in Matlab initially, we present a Software-In-The-Loop (SITL) simulation where the DOBC runs in a virtual environment which matches that of autopilot hardware running PX4. SITL simulation is an important step in the development of flight controllers since it allows for safe and efficient debugging and tuning. This is compared with most existing simulation results which are mostly Matlab-based (Antonelli et al., 2018; Moeini et al., 2019; Shao et al., 2018) and do not capture the many real-world constraints of on-board implementation (e.g. multi-rate measurements, control, and state estimation, bounded signals, and constrained hardware resources).

The proposed method is also implemented experimentally using the Applied Nonlinear Control Lab (ANCL) quadrotor platform in the presence of a wind disturbance. In simulation and experiment, the proposed method is compared with a traditional backstepping controller and the built-in PX4 motion controller. The results demonstrate improved performance for the proposed backstepping DOBC. It is important to note that our implementation does not require simplifying approximations (e.g. small-angle assumptions, negligible coupling between inner and outer loops) that are common in work with experimental validation (Antonelli et al., 2018; Dong et al., 2014; Shao et al., 2018). By basing our work on the open source and commonly used PX4 platform, our results are accessible to the research community. The self-contained PX4 code used in the paper is available online at Moeini and Lynch (2020) and can be used to reproduce our results. Other experimental work suffers from controller implementation on ground station computers (Cabecinhas et al., 2014, 2015). In this case, the wireless connection can limit the reliability and performance of the design.

The organisation of the paper is as follows: Section 2 presents the quadrotor modelling. In Section 3, the DOBC is presented and its stability proof given. In Section 4, the proposed method is validated in simulation and experiment.

2. Quadrotor dynamics

In this section, we review a traditional rigid body quadrotor model and establish notation for control design. The dynamics of the quadrotor has been presented in many works such as Castillo et al. (2005) and Bouabdallah (2007). We consider a traditional quadrotor UAV as shown in Figure 1. Two reference frames are needed: a fixed inertial navigation frame $\mathcal{N}$ with orthonormal basis $\{n_1, n_2, n_3\}$ and a body frame $\mathcal{B}$ whose origin is at the vehicle's centre of mass (CoM) and with orthonormal basis $\{b_1, b_2, b_3\}$. We define b_1 to point in the forward direction of vehicle, b_2 pointing right, and b_3 pointing down. The configuration of the quadrotor belongs to the special Euclidean group SE(3), and includes the position $p \in \mathbb{R}^3$ of the origin of $\mathcal{B}$ relative to $\mathcal{N}$, and the orientation $R \in \mathrm{SO}(3)$ of $\mathcal{B}$ with respect to $\mathcal{N}$. We assume each propeller generates thrust in the $-b_3$ direction and denote the total thrust created by all the propellers by the scalar input $u \geq 0$, i.e. the thrust vector is $-ub_3$. Controlling individual propeller speeds creates an input torque denoted $\tau \in \mathbb{R}^3$ which is expressed in $\mathcal{B}$. To simplify the presentation of the controller design, we take torque τ and thrust u as the system inputs. However, the actual physical inputs to the UAV

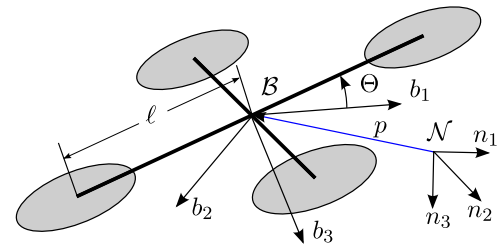

Figure 1. Diagram of a quadrotor showing body frame $\mathcal{B}$ and navigation frame $\mathcal{N}$.

propellers are PWM signals to the Electronic Speed Controller (ESC), which are denoted $W_i, i = 1, 2, 3, 4$. We assume a one-to-one relation between the torque and thrust (u, τ) and the physical inputs W_i, $i = 1, 2, 3, 4$. For a quadrotor in "cross" configuration we have

$$\begin{bmatrix} u \\ \tau \end{bmatrix} = \begin{bmatrix} K_u & K_u & K_u & K_u \\ -K_u \ell s_\Theta & K_u \ell s_\Theta & K_u \ell s_\Theta & -K_u \ell s_\Theta \\ K_u \ell c_\Theta & -K_u \ell c_\Theta & K_u \ell c_\Theta & -K_u \ell c_\Theta \\ -K_\tau & -K_\tau & K_\tau & K_\tau \end{bmatrix} \begin{bmatrix} \tilde{W}_1^2 \\ \tilde{W}_2^2 \\ \tilde{W}_3^2 \\ \tilde{W}_4^2 \end{bmatrix} \tag{1}$$

where K_u is a thrust constant, K_τ is a counter torque constant, ℓ is arm length, Θ is the angle between the arm and b_1 axis, $s_\Theta = \sin\Theta$, $c_\Theta = \cos\Theta$, and $\tilde{W}_i$ is a normalised PWM signal defined by $\tilde{W}_i = (W_i - W_{\min})/(W_{\max} - W_{\min})$ where $W_{\min}$ and $W_{\max}$ are the minimum and maximum values of W_i, respectively. Therefore, $\tilde{W}_i \in [0, 1]$ and the inputs (u and τ) are bounded. However, we do not explicitly account for this saturation in the control design. The multirotor dynamics can be expressed by

$$\dot{p} = v \tag{2a}$$

$$m\dot{v} = mgn_3 - uRn_3 + d_f \tag{2b}$$

$$\dot{R} = RS(\omega) \tag{2c}$$

$$J\dot{\omega} = -\omega \times J\omega + \tau + d_\tau \tag{2d}$$

where $v \in \mathbb{R}^3$ is linear velocity expressed in $\mathcal{N}$, $\omega \in \mathbb{R}^3$ is angular velocity expressed in $\mathcal{B}$, m is mass, J is inertia, g is the gravity constant, and $n_3 = [0, 0, 1]^\top$. The force disturbance $d_f \in \mathbb{R}^3$ and torque disturbance $d_\tau \in \mathbb{R}^3$ are unknown and used to model external disturbances such as wind gusts or other model uncertainties. The skew operator $S(\cdot) : \mathbb{R}^3 \to so(3)$ is defined as

$$S(x) = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix}, \quad \text{where } x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

In Section 3, we use ZYX Euler parameterisation for deriving the control for the yaw or heading angle. We denote the Euler angles as $\eta = [\phi, \theta, \psi]^\top$, where ϕ, θ and ψ are roll, pitch, and yaw, respectively. The rotational kinematics (2c) expressed in η are $\dot{\eta} = W(\eta)\omega$ with

$$W(\eta) = \begin{bmatrix} 1 & s_\phi t_\theta & c_\phi t_\theta \\ 0 & c_\phi & -s_\phi \\ 0 & s_\phi / c_\theta & c_\phi / c_\theta \end{bmatrix}$$

where $t_\theta = \tan\theta$.

3. Disturbance observer based control (DOBC) design

This section designs a trajectory tracking controller for the quadrotor dynamics (2). Given smooth bounded desired trajectories for position and yaw, denoted by $p_d \in \mathbb{R}^3$ and $\psi_d \in \mathbb{R}$, respectively, we derive a dynamic state feedback control for inputs u and τ to ensure exponential convergence of the tracking errors $p - p_d$ and $\psi - \psi_d$ in the presence of bounded constant disturbances d_f and d_τ. The full state measurement is assumed available. The proposed design adopts a backstepping approach inspired by Cabecinhas et al. (2014) and the references within. As discussed in the previous section, the approach in Cabecinhas et al. (2014) relies on six-dimensional parameter update laws to estimate d_f, whereas the proposed method uses one 3-dimensional DO for force disturbance estimation. Decoupling the observer from the state feedback ensures a minimal order of 3 for the DO of d_f. Since p and ψ can be independently controlled, the design is divided into two sections: position and yaw control.

3.1 Disturbance observer

We consider the DO for estimating the force disturbance

$$\hat{d}_f = z_{d_f} + k_{d_f} m v \tag{3a}$$

$$\dot{z}_{d_f} = -k_{d_f}\hat{d}_f - k_{d_f}(mgn_3 - uRn_3) \tag{3b}$$

where $z_{d_f} \in \mathbb{R}^3$ is the observer state, $\hat{d}_f \in \mathbb{R}^3$ is disturbance estimate, and $k_{d_f} > 0$ is an observer gain. Assuming d_f is constant (i.e. $\dot{d}_f = 0$) and defining $\tilde{d}_f = d_f - \hat{d}_f$, observer (3) has error dynamics

$$\begin{aligned}\dot{\tilde{d}}_f &= -(\dot{z}_{d_f} + k_{d_f} m\dot{v})\\ &= k_{d_f}\hat{d}_f + k_{d_f}(mgn_3 - uRn_3) - k_{d_f}(mgn_3 - uRn_3 + d_f)\\ &= -k_{d_f}\tilde{d}_f \end{aligned}\tag{4}$$

which is globally exponentially stable.

3.2 Position tracking control

In this section, we design a controller for position tracking. We start by defining a tracking error $\delta_1 = p - p_d$ and the first Lyapunov function candidate $V_1 = \frac{1}{2}\|\delta_1\|^2$. Using (2a) gives

$$\dot{V}_1 = \delta_1^\top \dot{\delta}_1 = \delta_1^\top (v - v_d) \tag{5}$$

where $v_d = \dot{p}_d$. Taking v as a virtual control to (5) we choose $\alpha_1 = v_d - k_1\delta_1$ as its desired value, where $k_1 > 0$ is a scalar controller gain. Defining the second error coordinate as $\delta_2 = mv - m\alpha_1$ we get

$$\dot{\delta}_1 = -k_1\delta_1 + \delta_2/m. \tag{6}$$

Substituting this expression into (5) we have $\dot{V}_1 = -k_1\|\delta_1\|^2 + \delta_1^\top\delta_2/m$. Next, considering the second Lyapunov function candidate $V_2 = V_1 + \frac{1}{2}\|\delta_2\|^2$ and taking its time derivative we have

$$\dot{V}_2 = -k_1\|\delta_1\|^2 + \frac{1}{m}\delta_2^\top\delta_1 + \delta_2^\top\dot{\delta}_2. \tag{7}$$

Substituting $d_f = \hat{d}_f + \tilde{d}_f$ and $\dot{\alpha}_1$ into the expression for $\dot{\delta}_2$ gives

$$\begin{aligned}\dot{\delta}_2 &= m\dot{v} - m\dot{\alpha}_1\\ &= mgn_3 - uRn_3 + \hat{d}_f + \tilde{d}_f - m\dot{v}_d + mk_1 v - mk_1 v_d\end{aligned}\tag{8}$$

where we have used (2b). Hence, substituting (8) into (7) gives

$$\begin{aligned}\dot{V}_2 = &-k_1\|\delta_1\|^2 + \frac{1}{m}\delta_2^\top\delta_1 + \delta_2^\top(mgn_3 - uRn_3 + \hat{d}_f + \tilde{d}_f\\ &- m\dot{v}_d + mk_1 v - mk_1 v_d).\end{aligned}\tag{9}$$

We denote α_2 as the desired value for the second virtual control uRn_3 and take

$$\alpha_2 = mgn_3 + \hat{d}_f - m\dot{v}_d + mk_1 v - mk_1 v_d + \frac{1}{m}\delta_1 + k_2\delta_2 \tag{10}$$

where $k_2 > 0$ is a controller gain. We remark that the virtual input uRn_3 is the thrust vector expressed in inertial frame $\mathcal{N}$. Introducing the third error coordinate $\delta_3 = \alpha_2 - uRn_3$ and substituting (10) into (9) gives

$$\dot{V}_2 = -k_1\|\delta_1\|^2 - k_2\|\delta_2\|^2 + \delta_2^\top\delta_3 + \tilde{d}_f^\top\delta_2. \tag{11}$$

and

$$\dot{\delta}_2 = -\delta_1/m - k_2\delta_2 + \delta_3 + \tilde{d}_f. \tag{12}$$

Now considering $V_3 = V_2 + \frac{1}{2}\|\delta_3\|^2$ as the third Lyapunov function candidate and taking its time derivative, by substituting (11) we get

$$\dot{V}_3 = -k_1\|\delta_1\|^2 - k_2\|\delta_2\|^2 + \delta_2^\top\delta_3 + \tilde{d}_f^\top\delta_2 + \delta_3^\top\dot{\delta}_3 \tag{13}$$

where

$$\begin{aligned}\dot{\delta}_3 &= \dot{\alpha}_2 - \dot{u}Rn_3 - uRS(\omega)n_3\\ &= \dot{\hat{d}}_f - m\ddot{v}_d + k_1(mgn_3 - uRn_3 + d_f) - mk_1\dot{v}_d\\ &\quad + \frac{1}{m}(v - v_d) + k_2(mgn_3 - uRn_3 + d_f\\ &\quad - m\dot{v}_d + mk_1 v - mk_1 v_d) - \dot{u}Rn_3 - uRS(\omega)n_3.\end{aligned}\tag{14}$$

Substituting (4) and $d_f = \hat{d}_f + \tilde{d}_f$ in (14) we obtain

$$\dot{\delta}_3 = \beta + (k_{d_f} + k_1 + k_2)\tilde{d}_f - \dot{u}Rn_3 - uRS(\omega)n_3 \tag{15}$$

where β is a function of known variables

$$\begin{aligned}\beta = &-m\ddot{v}_d + k_1 mgn_3 - k_1 uRn_3 + k_1\hat{d}_f - mk_1\dot{v}_d\\ &+ \frac{1}{m}v - \frac{1}{m}v_d + k_2 mgn_3 - k_2 uRn_3 + k_2\hat{d}_f\\ &- k_2 m\dot{v}_d + k_2 mk_1 v - k_2 mk_1 v_d.\end{aligned}$$

Hence, substituting (15) into (13) gives

$$\begin{aligned}\dot{V}_3 = &-k_1\|\delta_1\|^2 - k_2\|\delta_2\|^2 + \delta_2^\top\delta_3\\ &+ \delta_3^\top(\beta - \dot{u}Rn_3 - uRS(\omega)n_3)\\ &+ \tilde{d}_f^\top(\delta_2 + (k_{d_f} + k_1 + k_2)\delta_3)\end{aligned}$$

The terms $uRS(\omega)n_3$ and $\dot{u}Rn_3$ can be written as

$$uRS(\omega)n_3 = R\begin{bmatrix} u\omega_2 \\ -u\omega_1 \\ 0 \end{bmatrix}, \quad \dot{u}Rn_3 = R\begin{bmatrix} 0 \\ 0 \\ \dot{u} \end{bmatrix}.$$

Since u is the system input, we can assign the value of $\dot{u}$ at this stage and continue the backstepping design with $uRS(\omega)n_3$ as a virtual control. So we choose

$$\dot{u} = n_3^\top R^\top(\beta + \delta_2 + k_3\delta_3) \tag{16}$$

where $k_3 > 0$ is a controller gain. Letting $\delta_4 = \alpha_3 - uRS(\omega)n_3$ where

$$\alpha_3 = R[I - n_3n_3^\top]R^\top(\beta + \delta_2 + k_3\delta_3) \tag{17}$$

gives us

$$\dot{\delta}_3 = -\delta_2 - k_3\delta_3 + \delta_4 + (k_{d_f} + k_1 + k_2)\tilde{d}_f. \tag{18}$$

The choice (16) and (17) cancels the effect of indefinite terms $\delta_2^\top\delta_3$ and $\delta_3^\top\beta$ in $\dot{V}_3$ and adds a damping term $-k_3\|\delta_3\|^2$. We have

$$\begin{aligned}\dot{V}_3 = &-k_1\|\delta_1\|^2 - k_2\|\delta_2\|^2 - k_3\|\delta_3\|^2 + \delta_3^\top\delta_4 \\ &+ \tilde{d}_f^\top(\delta_2 + (k_{d_f} + k_1 + k_2)\delta_3).\end{aligned}$$

Considering $V_4 = V_3 + \frac{1}{2}\|\delta_4\|^2$ as a new Lyapunov function candidate, we have

$$\begin{aligned}\dot{V}_4 = &-k_1\|\delta_1\|^2 - k_2\|\delta_2\|^2 - k_3\|\delta_3\|^2 + \delta_3^\top\delta_4 \\ &+ \tilde{d}_f^\top(\delta_2 + (k_{d_f} + k_1 + k_2)\delta_3) + \delta_4^\top\dot{\delta}_4\end{aligned}$$

where

$$\dot{\delta}_4 = \dot{\alpha}_3 - \dot{u}RS(\omega)n_3 - uRS(\omega)^2n_3 - uRS(\dot{\omega})n_3 \tag{19}$$

and

$$\begin{aligned}\dot{\alpha}_3 = &RS(\omega)[I - n_3n_3^\top]R^\top(\beta + \delta_2 + k_3\delta_3) \\ &+ R[I - n_3n_3^\top]S(\omega)^\top R^\top(\beta + \delta_2 + k_3\delta_3) \\ &+ R[I - n_3n_3^\top]R^\top(\dot{\beta} + \dot{\delta}_2 + k_3\dot{\delta}_3).\end{aligned}$$

Calculating expressions for $\dot{\beta}$, $\dot{\delta}_2$ and $\dot{\delta}_3$ and substituting $d_f = \hat{d}_f + \tilde{d}_f$ we obtain

$$\begin{aligned}\dot{\beta} &= \dot{\beta}' + \left(k_1k_{d_f} + \frac{1}{m^2} + k_2k_{d_f} + k_1k_2\right)\tilde{d}_f \\ \dot{\delta}_2 &= \dot{\delta}_2' + \tilde{d}_f \\ \dot{\delta}_3 &= \dot{\delta}_3' + (k_{d_f} + k_1 + k_2)\tilde{d}_f\end{aligned}$$

where $\dot{\beta}'$, $\dot{\delta}_2'$ and $\dot{\delta}_3'$ are the known parts of $\dot{\beta}$, $\dot{\delta}_2$ and $\dot{\delta}_3$, respectively, and given by

$$\begin{aligned}\dot{\beta}' = &-m\dddot{v}_d - k_1\dot{u}Rn_3 - k_1uRS(\omega)n_3 - mk_1\ddot{v}_d \\ &+ \frac{1}{m^2}(mgn_3 - uRn_3 + \hat{d}_f) - \frac{1}{m}\dot{v}_d \\ &- k_2\dot{u}Rn_3 - k_2uRS(\omega)n_3 - k_2m\ddot{v}_d \\ &+ k_2k_1(mgn_3 - uRn_3 + \hat{d}_f) - k_1k_2m\dot{v}_d\end{aligned}$$

$$\begin{aligned}\dot{\delta}_2' &= mgn_3 - uRn_3 + \hat{d}_f - m\dot{v}_d + mk_1v - mk_1v_d \\ \dot{\delta}_3' &= \beta - \dot{u}Rn_3 - uRS(\omega)n_3.\end{aligned}$$

Therefore, the known part of $\dot{\alpha}_3$ can be defined by $\dot{\alpha}_3' : \dot{\alpha}_3 = \dot{\alpha}_3' + kR[I - n_3n_3^\top]R^\top\tilde{d}_f$ where

$$\begin{aligned}\dot{\alpha}_3' = &RS(\omega)[I - n_3n_3^\top]R^\top(\beta + \delta_2 + k_3\delta_3) \\ &+ R[I - n_3n_3^\top]S(\omega)^\top R^\top(\beta + \delta_2 + k_3\delta_3) \\ &+ R[I - n_3n_3^\top]R^\top(\dot{\beta}' + \dot{\delta}_2' + k_3\dot{\delta}_3')\end{aligned}$$

and

$$k = \left(k_1k_{d_f} + \frac{1}{m^2} + k_2k_{d_f} + k_1k_2\right) + 1 + k_3(k_{d_f} + k_1 + k_2).$$

Now we can write the expression for $\dot{V}_4$ as

$$\begin{aligned}\dot{V}_4 = &-k_1\|\delta_1\|^2 - k_2\|\delta_2\|^2 - k_3\|\delta_3\|^2 \\ &+ \delta_4^\top(\dot{\alpha}_3' + \delta_3 - \dot{u}RS(\omega)n_3 - uRS(\omega)^2n_3 - uRS(\dot{\omega})n_3) \\ &+ \tilde{d}_f^\top(\delta_2 + (k_{d_f} + k_1 + k_2)\delta_3 + kR[I - n_3n_3^\top]R^\top\delta_4)\end{aligned}$$

At this stage, we define a desired value for the rate of angular velocity $\dot{\omega}_d$, such that

$$\begin{aligned}&\delta_4^\top(\dot{\alpha}_3' + \delta_3 - \dot{u}RS(\omega)n_3 - uRS(\omega)^2n_3 - uRS(\dot{\omega}_d)n_3) \\ &\quad = -k_4\delta_4^\top\delta_4\end{aligned}$$

where $k_4 > 0$ is a controller gain. This leads to the expression $-k_4\|\delta_4\|^2$ appearing in $\dot{V}_4$. Considering the structure for $S(\dot{\omega}_d)n_3$, we can write the above equation as

$$\begin{aligned}\delta_4^\top R\begin{bmatrix} \dot{\omega}_{d2} \\ -\dot{\omega}_{d1} \\ 0 \end{bmatrix}u = &\delta_4^\top(\dot{\alpha}_3' + \delta_3 - \dot{u}RS(\omega)n_3 \\ &- uRS(\omega)^2n_3 + k_4\delta_4).\end{aligned} \tag{20}$$

Now using the fact that the third component of $R^\top\delta_4$ is zero, we conclude that (20) is satified by selecting

$$\dot{\omega}_{d1} = -\frac{n_2^\top R^\top}{u}(\dot{\alpha}_3' - \dot{u}RS(\omega)n_3 - uRS(\omega)^2n_3 + \delta_3 + k_4\delta_4) \tag{21a}$$

$$\dot{\omega}_{d2} = \frac{n_1^\top R^\top}{u}(\dot{\alpha}_3' - \dot{u}RS(\omega)n_3 - uRS(\omega)^2n_3 + \delta_3 + k_4\delta_4). \tag{21b}$$

In order to obtain the expressions for the torque input τ which achieves position tracking, we consider $\dot{\omega}_{d3} = 0$. Now we try to get the expression for torque which is the actual sytem input.

From the rotational dynamics, we consider

$$\tau = J\dot{\omega}_d + \omega \times J\omega - \hat{d}_\tau \tag{22}$$

where, $\hat{d}_\tau$ is the torque disturbance estimate introduced to cancel the effect of d_τ. From (22) and (2d), we get

$$\dot{\omega} = \dot{\omega}_d + J^{-1}\tilde{d}_\tau. \tag{23}$$

Substituting (21) and (23) in $\dot{\delta}_4$, gives us

$$\dot{\delta}_4 = -\delta_3 - k_4\delta_4 + uRS(n_3)J^{-1}\tilde{d}_\tau + kR[I - n_3n_3^\top]R^\top\tilde{d}_f. \tag{24}$$

Now we can write the following expression for $\dot{V}_4$

$$\begin{aligned}\dot{V}_4 = &-k_1\|\delta_1\|^2 - k_2\|\delta_2\|^2 - k_3\|\delta_3\|^2 - k_4\|\delta_4\|^2 \\ &+ \delta_4^\top uRS(n_3)J^{-1}\tilde{d}_\tau + \tilde{d}_f^\top(\delta_2 + (k_{d_f} + k_1 + k_2)\delta_3 \\ &+ kR[I - n_3n_3^\top]R^\top\delta_4).\end{aligned}$$

Therefore, at this stage, we design a torque disturbance observer using the same observer structure we used for the force disturbance

$$\hat{d}_\tau = z_{d_\tau} + k_{d_\tau}J\omega \tag{25a}$$

$$\dot{z}_{d_\tau} = -k_{d_\tau}\hat{d}_\tau - k_{d_\tau}(-\omega \times J\omega + \tau) \tag{25b}$$

which results in exponentially stable error dynamics $\dot{\tilde{d}}_\tau = -k_{d_\tau}\tilde{d}_\tau$, where $k_{d_\tau} > 0$.

The closed-loop stability of the proposed backstepping controller is summarised by the following theorem.

Theorem 3.1: *Given system* (2) *with constant disturbances* d_f *and* d_τ, *bounded smooth reference trajectory* p_d *and assuming* $u > 0$, *the equilibrium* $[\delta_1^\top, \delta_2^\top, \delta_3^\top, \delta_4^\top, \tilde{d}_f^\top, \tilde{d}_\tau^\top]^\top = 0$ *of the closed-loop dynamics is exponentially stable with dynamic state feedback control* (3), (16), (21), (22) *and* (25).

Proof: Denoting $\delta = [\delta_1^\top, \delta_2^\top, \delta_3^\top, \delta_4^\top]^\top$ and $\tilde{d} = [\tilde{d}_f^\top, \tilde{d}_\tau^\top]^\top$ then from (6), (12), (18) and (24) the tracking error dynamics is

$$\dot{\delta} = A\delta + B(t)\tilde{d} \tag{26}$$

where

$$A = \begin{bmatrix} -k_1I_3 & I_3/m & 0_3 & 0_3 \\ -I_3/m & -k_2I_3 & I_3 & 0_3 \\ 0_3 & -I_3 & -k_3I_3 & I_3 \\ 0_3 & 0_3 & -I_3 & -k_4I_3 \end{bmatrix},$$

$$B(t) = \begin{bmatrix} 0_3 & 0_3 \\ I_3 & 0_3 \\ (k_{d_f} + k_1 + k_2)I_3 & 0_3 \\ kR[I - n_3n_3^\top]R^\top & uRS(n_3)J^{-1} \end{bmatrix}.$$

where I_3 denotes the 3×3 identity matrix and 0_3 denotes the 3×3 zero matrix. Using the result (Marino & Tomei, 1995, Lemma. III.1), we can show the equilibrium of (26) is exponentially stable if its zero-input dynamics is exponentially stable, $\|B(t)\| < B_M$ for some constant B_M, and input $\tilde{d}$ is exponentially convergent. Since the eigenvalues of A have negative real parts for any $k_1, k_2, k_3, k_4 > 0$, we know the zero-input dynamics of (26) is exponentially stable. Also, $\|B(t)\|$ is bounded since the rotation matrix R and thrust u are bounded. Since the disturbance estimate error dynamics is exponentially stable, we know $\tilde{d}$ is exponentially convergent, i.e. $\|\tilde{d}\| \le Me^{-\alpha t}$, where $\alpha, M > 0$. Therefore, the equilibrium $[\delta_1^\top, \delta_2^\top, \delta_3^\top, \delta_4^\top, \tilde{d}_f^\top, \tilde{d}_\tau^\top]^\top = 0$ of closed-loop (2), (3), (16), (21), (22), and (25) is exponentially stable. ■

3.3 Yaw and position tracking control

In this subsection, we develop a controller to track trajectories for yaw using τ_3. Given a smooth bounded reference trajectory ψ_d, we define the tracking error $\epsilon_1 = \psi - \psi_d$, and consider the Lyapunov function candidate $V_{\psi_1} = \frac{1}{2}\|\epsilon_1\|^2$ whose time derivative is $\dot{V}_{\psi 1} = \epsilon_1(\dot{\psi} - \dot{\psi}_d)$. We take $\dot{\psi}$ as a virtual control and define $\epsilon_2 = \dot{\psi} - \alpha_{\psi 1}$, where $\alpha_{\psi 1} = \dot{\psi}_d - k_{\psi 1}\epsilon_1$ is the desired value for $\dot{\psi}$, and $k_{\psi 1} > 0$ is a controller gain. We obtain $\dot{V}_{\psi_1} = -k_{\psi 1}\|\epsilon_1\|^2 + \epsilon_1\epsilon_2$. Defining the second Lyapunov function candidate as $V_{\psi 2} = V_{\psi 1} + \frac{1}{2}\|\epsilon_2\|^2$, we get

$$\dot{V}_{\psi_2} = \dot{V}_{\psi_1} + \epsilon_2\dot{\epsilon}_2 = -k_{\psi 1}\|\epsilon_1\|^2 + \epsilon_1\epsilon_2 + \epsilon_2(\ddot{\psi} - \dot{\alpha}_{\psi 1}) \tag{27}$$

where $\dot{\alpha}_{\psi 1} = \ddot{\psi}_d - k_{\psi 1}(\dot{\psi} - \dot{\psi}_d)$. At this stage, we can immediately assign the value of $\ddot{\psi}$ since it is directly related to $\dot{\omega}$ which can be considered as the system input. Next, we present the relation between $\ddot{\psi}$ and $\dot{\omega}$. Taking time derivative of $\dot{\eta} = W(\eta)\omega$ gives $\ddot{\eta} = \dot{W}(\eta)\omega + W(\eta)\dot{\omega}$. From the last row of this equation, we have

$$\ddot{\psi} = n_3^\top\dot{W}(\eta)\omega + \frac{s_\phi}{c_\theta}\dot{\omega}_2 + \frac{c_\phi}{c_\theta}\dot{\omega}_3. \tag{28}$$

Since $\dot{\omega}_2$ and $\dot{\omega}_3$ are not measurable, we replace them by their desired values using (23):

$$\ddot{\psi} = n_3^\top\dot{W}(\eta)\omega + \frac{s_\phi}{c_\theta}\dot{\omega}_{d2} + \frac{c_\phi}{c_\theta}\dot{\omega}_{d3} + \frac{s_\phi}{J_{22}c_\theta}\tilde{d}_{\tau 2} + \frac{c_\phi}{J_{33}c_\theta}\tilde{d}_{\tau 3} \tag{29}$$

where $\tilde{d}_\tau = [\tilde{d}_{\tau 1}, \tilde{d}_{\tau 2}, \tilde{d}_{\tau 3}]^\top$, and J_{ij} is the (i,j)th entry of J. In order to simplify (29) and without loss of generality, we have considered the practical case of J diagonal. Now, if we consider $\dot{\omega}_{d3}$ as

$$\dot{\omega}_{d3} = \frac{c_\theta}{c_\phi}\left(\dot{\alpha}_{\psi 1} - \epsilon_1 - k_{\psi 2}\epsilon_2 - n_3^\top\dot{W}(\eta)\omega - \frac{s_\phi}{c_\theta}\dot{\omega}_{d2}\right) \tag{30}$$

and substitute (30) into (28), we obtain

$$\ddot{\psi} = \dot{\alpha}_{\psi 1} - \epsilon_1 - k_{\psi 2}\epsilon_2 + \frac{s_\phi}{J_{22}c_\theta}\tilde{d}_{\tau 2} + \frac{c_\phi}{J_{33}c_\theta}\tilde{d}_{\tau 3}. \tag{31}$$

We observe (30) has singular points at $\phi = \pi/2 + k\pi, k \in \mathbb{Z}$. Further, from the definition of $W(\eta)$, $\theta = \pi/2 + k\pi, k \in \mathbb{Z}$ are singular points for the Euler angles. We must therefore assume $(\phi, \theta) \in \mathcal{S} = \{(\phi, \theta) : -\pi/2 < \phi < \pi/2, -\pi/2 < \theta < \pi/2\}$ when controlling yaw and position. Considering the Lyapunov function candidate $V_5 = V_4 + V_{\psi_2}$ and substituting (31) into (27) results

$$\dot{V}_5 = -k_1\|\delta_1\|^2 - k_2\|\delta_2\|^2 - k_3\|\delta_3\|^2 - k_4\|\delta_4\|^2 - k_{\psi 1}\|\epsilon_1\|^2$$

$$- k_{\psi 2}\|\epsilon_2\|^2 + \tilde{d}_\tau^\top (uJ^{-1}S(n_3)^\top R^\top \delta_4 + G\epsilon_2)$$
$$+ \tilde{d}_f^\top (\delta_2 + (k_{d_f} + k_1 + k_2)\delta_3 + kR[I - n_3 n_3^\top]R^\top \delta_4)$$

where $G = [0, \frac{s_\phi}{J_{22}c_\theta}, \frac{c_\phi}{J_{33}c_\theta}]^\top$.

Theorem 3.2: *Given system* (2) *with constant disturbances* d_f *and* d_τ *and bounded smooth reference trajectories* p_d, ψ_d, *assuming* $u > 0$ *and provided* $(\phi, \theta) \in \mathcal{S}$, *the equilibrium* $[\delta_1^\top, \delta_2^\top, \delta_3^\top, \delta_4^\top, \epsilon_1, \epsilon_2, \tilde{d}_f^\top, \tilde{d}_\tau^\top]^\top = 0$ *of the closed-loop system is exponentially stable with dynamic state feedback control* (3), (16), (21), (22), (25) *and* (30).

Proof: The procedure of the proof is similar to the proof of Theorem 3.1 except matrices A and B are

$$A = \begin{bmatrix} -k_1 I_3 & I_3/m & 0_3 & 0_3 & 0_{3\times 1} & 0_{3\times 1} \\ -I_3/m & -k_2 I_3 & I_3 & 0_3 & 0_{3\times 1} & 0_{3\times 1} \\ 0_3 & -I_3 & -k_3 I_3 & I_3 & 0_{3\times 1} & 0_{3\times 1} \\ 0_3 & 0_3 & -I_3 & -k_4 I_3 & 0_{3\times 1} & 0_{3\times 1} \\ 0_{1\times 3} & 0_{1\times 3} & 0_{1\times 3} & 0_{1\times 3} & -k_{\psi 1} & 1 \\ 0_{1\times 3} & 0_{1\times 3} & 0_{1\times 3} & 0_{1\times 3} & -1 & -k_{\psi 2} \end{bmatrix},$$

$$B(t) = \begin{bmatrix} 0_3 & 0_3 \\ I_3 & 0_3 \\ (k_{d_f} + k_1 + k_2)I_3 & 0_3 \\ kR(I - n_3 n_3^\top)R^\top & uRS(n_3)J^{-1} \\ 0_{1\times 3} & 0_{1\times 3} \\ 0_{1\times 3} & G^\top \end{bmatrix}.$$

■

An important contribution of this work is the decoupled structure of the control law. The DO design is decoupled from the state feedback control and this simplifies the controller expressions and allows for implementation on a commonly used autopilot such as PX4/Pixhawk. Also, assuming $\tilde{d}$ is an input the closed-loop tracking error dynamics is linear, and exponentially stability can be readily proven. This should be compared to the adaptive backstepping control in Cabecinhas et al. (2014) which achieves asymptotic stability. In fact, most of the mentioned existing work in Section 1 only provides asymptotic stability results, sometimes only for an outer loop subsystem (Shao et al., 2018) or for approximate models (Dong et al., 2014; Lee et al., 2016).

4. Experimental validation

Simulation and experimental flight test of the proposed control law are necessary for validating its performance. This testing demonstrates if the design can tolerate unmodelled effects such as measurement noise and input saturation under typical operating conditions. In this section, we present SITL simulations and experimental flight testing results.

Table 1. ANCL-Q3 model parameters.

J_1	J_2	J_3	m	ℓ	Θ	K_u	K_τ
0.03 kg^2	0.03 kg^2	0.05 kg^2	1.6 kg	0.165 m	1.00 rad	7.75 N	0.8 N m

4.1 Hardware platform

The hardware components of the ANCL quadrotor platform consist of a quadrotor vehicle ANCL-Q3, a Vicon MCS, and a QGroundControl ground station (QGC).

4.1.1 The quadrotor vehicle ANCL-Q3

The ANCL-Q3 quadrotor is shown in Figure 2 which consists of a 3D Robotics quadrotor do-it-yourself (DIY) frame in a cross configuration equipped with a Pixhawk 1 flight controller which has a 180 MHz ARM CPU, two 3D accelerometers, two 3D gyroscopes, a 3D magnetometer, and a pressure sensor (Meier et al., 2012). ANCL-Q3 includes a 2.4 GHz LairdTech transceiver connected to the Pixhawk to receive data from the MCS. The Pixhawk is also connected to a Spektrum satellite receiver which is paired to a Spektrum DX8 transmitter. The DX8 enables manual control and allows the operator to switch between different control modes. A RN-XV WiFly Module is used to connect the PX4 to the local network via WiFi. ANCL-Q3 is powered by a 12 V, 3 cell, 5000 mAh lithium polymer battery (LiPo). This provides a flight time of about 10 minutes. The Pixhawk outputs a PWM signal to Afro 30 A ESCs which are connected to Turnigy 1100 KV Brushless Outrunner Motors. The diameters of the APC propellers are 11". The Pixhawk logs data onto its SD card. The mass of the quadrotor is 1.6 kg including the battery. The quadrotor is designed

Figure 2. The ANCL-Q3 quadrotor vehicle.

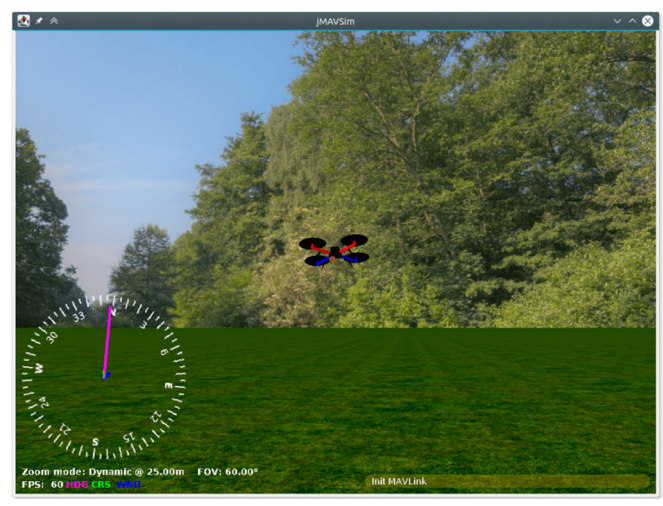

Figure 3. Visualizing the simulated motion control in jMAVSim.

such that it hovers at approximately 50% of its maximum thrust. This ensures enough thrust is available when aggressive manoeuvres are required. Table 1 lists model parameters in (2) and (1).

4.1.2 The motion capture system (MCS)

The MCS uses a network of eight Vicon Bonita 3 (B3) cameras, a Windows PC, and a 2.4 GHz LairdTech transceiver paired to the one on-board ANCL-Q3. The B3 is a 0.3 MP near infrared (NIR) camera with maximum frame rate of 240 frames per second (fps) and a resolution of 640 × 480. The quadrotor can be detected by using four 38.1 mm reflective markers mounted on its frame. The position and yaw of the UAV are estimated by the MCS which introduces a constant 10 ms delay (Fink, 2018). Linear velocity is estimated using a low-pass filtered finite difference of position. The data rate of LairdTech transceiver is about 100 Hz and includes a constant 15 ms delay (Fink, 2018).

4.1.3 The ground station

The quadrotor communicates to a ground control station running QGC (Meier, 2017) that is an open-source ground control

Table 2. Root mean square error (RMSE) of the steady-state position tracking error in the SITL/jMAVSim simulation.

Controller	PX4MC	BS	BSDOBC
RMSE($\|\tilde{p}\|$) m	0.9583	0.2907	0.0268

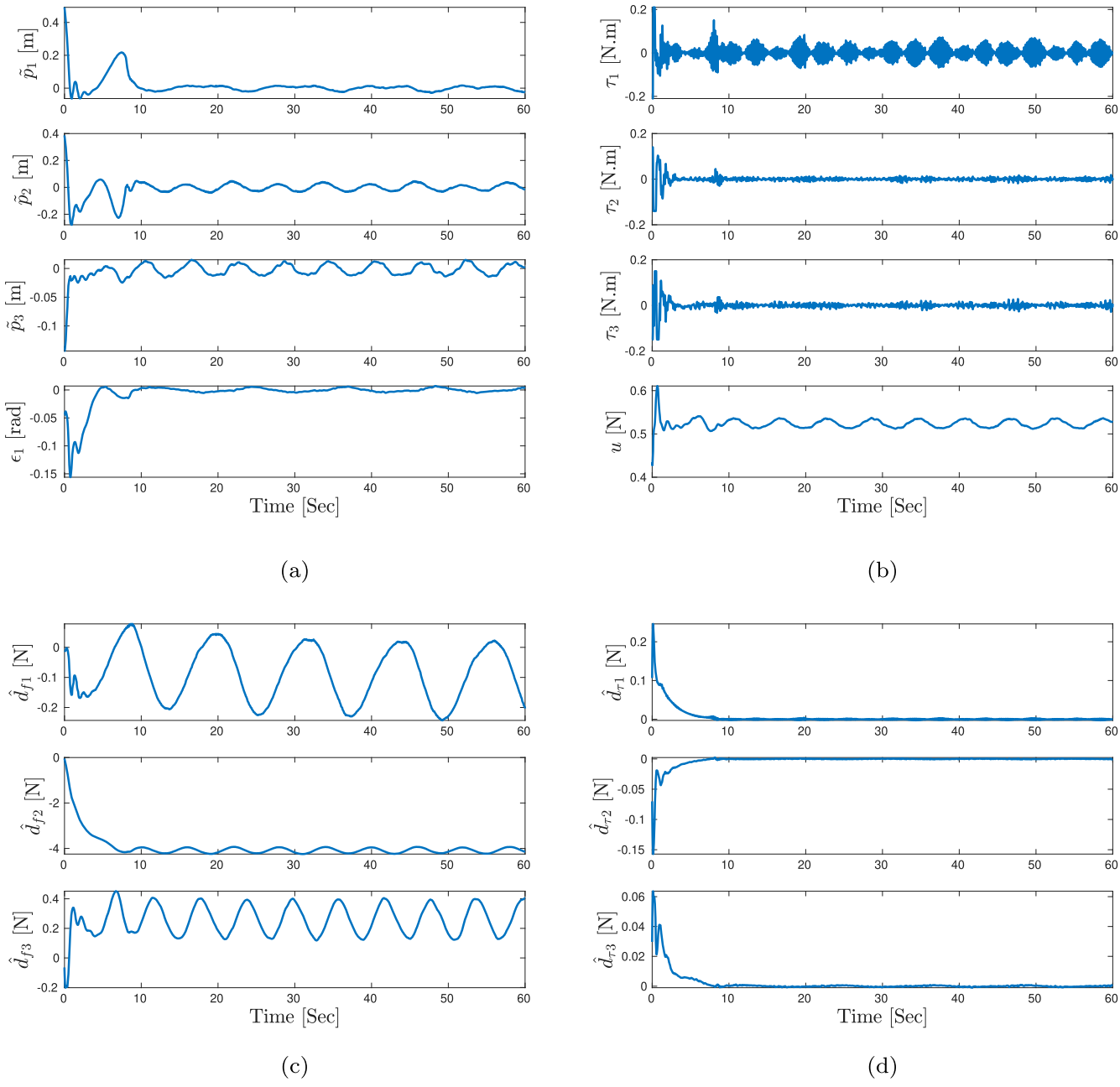

Figure 4. Trajectory tracking error, controller inputs, force and torque disturbance estimates for the BSDOBC in the SITL/jMAVSim simulation. (a) Tracking error. (b) Normalized controller inputs. (b) Force disturbance estimates and (c) Torque disturbance estimates.

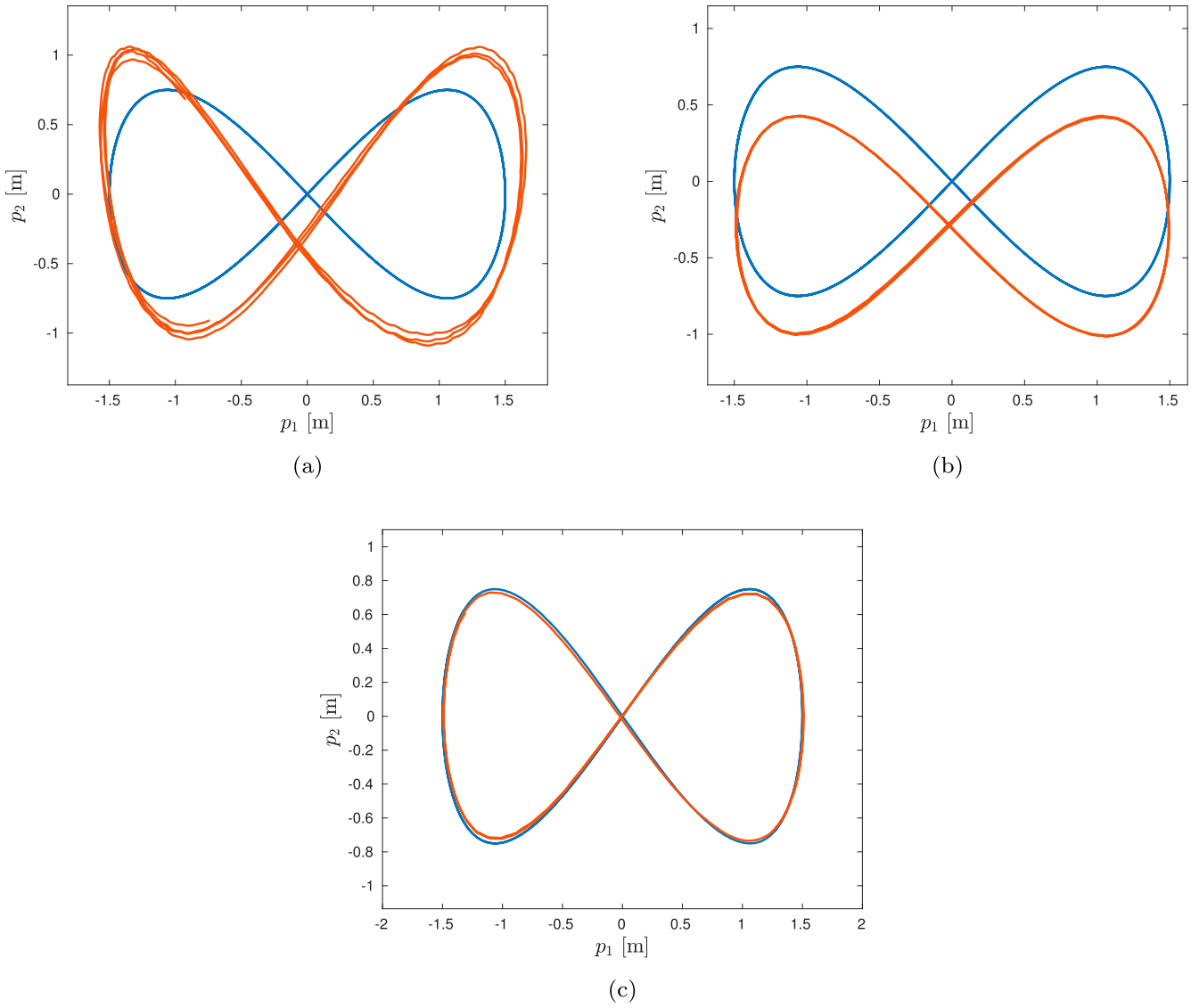

Figure 5. Desired and actual trajectories for the control methods simulated using SITL/jMAVSim. (a) PX4MC. (b) BS and (c) BSDOBC.

software and can be used for flashing the Pixhawk, parameter tuning, data visualisation, status monitoring, and mission planning. QGC and PX4 are connected through WiFi. The MAVLink protocol (Meier, 2015) is used for communication from the MCS to ANCL-Q3 and between ANCL-Q3 and QGC.

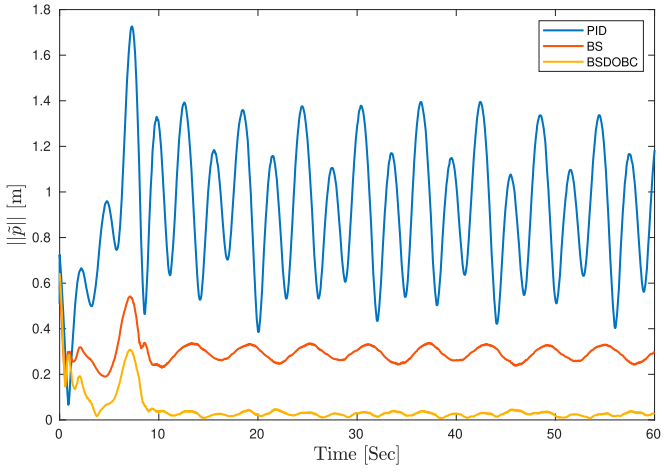

Figure 6. Norm of trajectory tracking error simulated using SITL/jMAVSim.

4.2 On-board controller implementation

We implement our proposed control on the PX4 autopilot firmware which is an active and mature open-source project intended for research and commercial applications (Meier, 2018). PX4 runs on a number of hardware platforms and various robot types. It has a modular structure where individual autopilot functionality (e.g. motion control, state estimation) is separated into self-contained modules which each runs as a task on NuttX RTOS platforms (e.g. Pixhawk) or as a thread within the main PX4 process on POSIX platforms

Table 3. Observer and controller gains used in experiment.

k_1	k_2	k_3	k_4	$k_{\psi 1}$	$k_{\psi 2}$	k_{d_f}
14	8.5	1.5	1.5	3.1	1.3	2.7

(e.g. SITL). This modular structure has a number of advantages. For example, adding a new control method to the system is relatively straightforward as only a small component of the code with a well-defined interface needs to be modified. The so-called PX4 middleware supports communication between the modules, sensor device drivers, and communication outside PX4. Module communication is implemented with the micro-object request broker (uORB) which provides a publish/subscribe bus. Using uORB, publishers send messages (e.g. a UAV control input) onto a bus instead of sending the messages directly to specific subscriber modules. Also, subscribers receive messages as soon as there are updates. PX4 provides driver modules for the hardware components (e.g. IMU, GPS and PWM outputs) and libraries for programming (e.g. matrix computation).

To implement the backstepping DOBC we created a new module using the v1.5.5 release of PX4. To simplify its implementation the module uses a so-called "Block" structure in the Controller Library contained in `px4/src/lib/controllib`. This library simplifies the process of subscription and publication. The library also provides functions for numerical integration which are used in the disturbance observer (3) and integration for thrust in (16). The controller module also uses the PX4 Matrix Library (`https://github.com/PX4/Matrix`) to improve the readability of the control law expressions given in Section 3. Control gains are implemented using PX4's parameter system. This simplifies controller tuning by allowing users to adjust gains from the PXH command line or from QGC. Roll, pitch and angular velocity are estimated onboard by an AHRS module `attitude_estimator_q` which uses the Pixhawk IMU measurements. Position and linear velocity are obtained from `local_position_estimator` which uses the MCS and IMU data. Further details on the modules used are in Fink (2018).

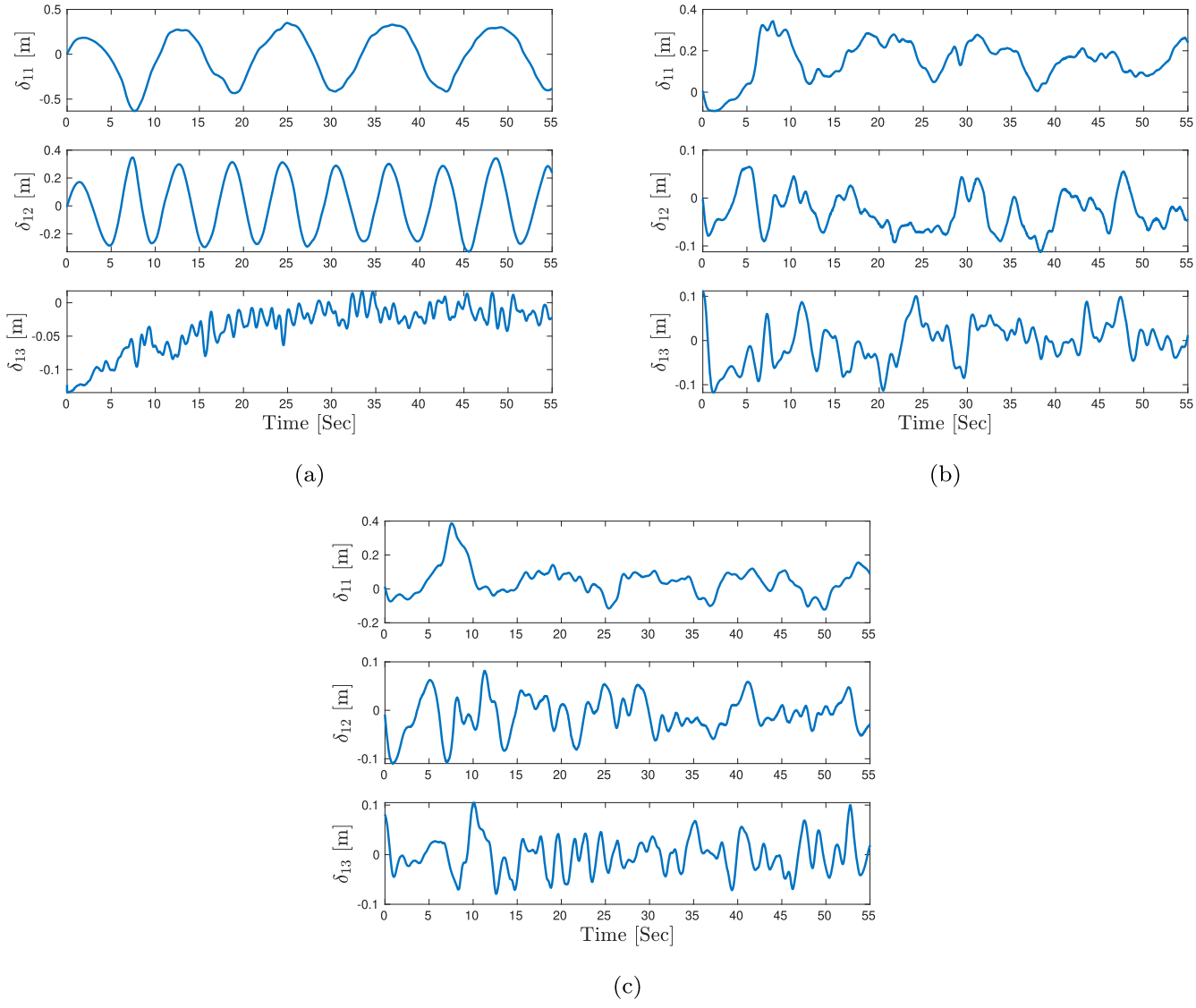

Figure 7. Experimental results: tracking error. (a) PX4MC. (b) BS and (c) BSDOBC.

4.3 SITL jMAVSim simulation

Performing a simulation that recreates the actual flight conditions and matches on-board implementation has several advantages when developing motion control algorithms. SITL simulation is a method that allows you to run the actual autopilot code and investigate its behaviour without any hardware. In this method, the code is running on a PC and is interfaced with a simulator that models the vehicle dynamics and the environment conditions (e.g. wind-gust). Therefore, SITL simulation can be an important step to the actual flight testing, by accelerating the debugging process and controller tuning.

In this section, we present SITL simulation results by employing the PX4 SITL framework and the jMAVSim simulator. SITL and jMAVSim are built into the PX4 project. The jMAVSim simulator receives PWM inputs from the autopilot code and outputs GPS position, IMU and barometer measurements related to the simulated UAV motion.

jMAVSim is developed in Java and employs `java3d` library for visualisation of a 3D virtual flight environment. Figure 3 shows a screenshot of this visualisation. Communication between PX4 and jMAVSim is done by the MAVLink protocol using localhost User Datagram Protocol (UDP). A PXH shell, similar to NSH shell in the Pixhawk/NuttX platform, facilitates code debugging and controlling the modules. It can be used to set the parameters, start and stop the modules and listen to the PX4 messages.

The UAV's model parameters can be hardcoded in the `Simulator` class. To match the real experiment presented in Section 4, we chose the model parameters as specified in Table 1 with a cross (or "x") vehicle configuration. The jMAVSim simulator accepts normalised torque and thrust inputs from PX4 and therefore includes thrust saturation. To model the actuator dynamics, the normalised inputs are low-pass filtered to generate individual rotor thrust (see class `Rotor`). The time constant of the low-pass filter is set to 5 ms. To recreate the actual flight conditions, jMAVSim adds adjustable time delay to GPS measurements and zero mean Gaussian noise to the gyro, accelerometer and magnetometer measurements. The default numbers for these parameters were chosen. The

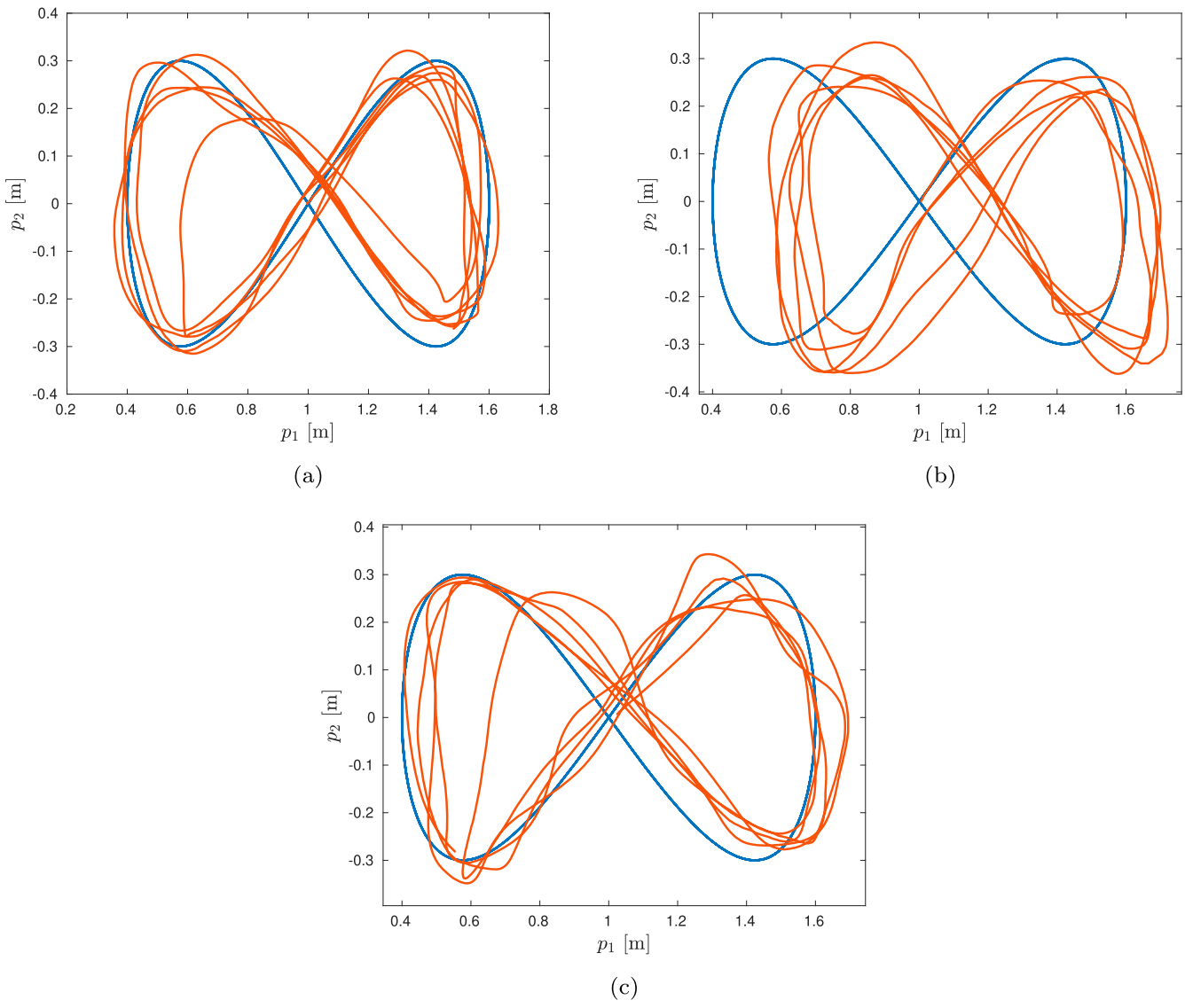

Figure 8. Experimental results: p_1-p_2 plot. (a) PX4MC. (b) BS and (c) BSDOBC.

thrust constant is scaled such that normalised thrust is 0.51 in hover.

$$u_n = 0.51 + \frac{u - mg}{60} \tag{32}$$

This value was obtained from the stock PX4 controller `mc_pos_control` when the quadrotor is in hover. The torque commands τ_1, τ_2 and τ_3 are also normalised by factors $n_{\tau_1} = 0.33$, $n_{\tau_2} = 0.33$ and $n_{\tau_3} = 1$, respectively.

Environmental disturbances can be easily generated in jMAVSim. We use this feature to create the force disturbance d_f. The model that generates the force disturbance is in the `AbstractMultiCopter` class and the force disturbance it outputs is proportional to the airspeed i.e.

$$d_f = c \cdot (w - v) \tag{33}$$

where c is a drag coefficient whose default value 0.03 Ns/m is used and $w \in \mathbb{R}^3$ is the wind velocity vector expressed in $\mathcal{N}$. The wind velocity is a random process whose variance and mean can be controlled by the user and is obtained from

$$\frac{\mathrm{d}w}{\mathrm{d}t} = -\frac{w}{\tau} + \left(\frac{W}{\tau} + \nu\right) \tag{34}$$

where we take $W = [0, -20, 0]^T$ m/s as the constant which determines the mean steady-state value of w, $\tau = 2$ s and $\nu = [\nu_1, \nu_2, \nu_3]^T \in \mathbb{R}^3$ where its components are Gaussian white noise processes with zero mean and variance $(\tau\sigma_{wk})^2$, i.e. $\mathcal{N}(0, (\tau\sigma_{wk})^2)$, where $\sigma_{w1} = 6$ m/s, $\sigma_{w2} = 8$ m/s, $\sigma_{w3} = 0$. The parameters σ_{wk} and W can be set using the GUI or can be coded in the `Simulator` class. To investigate the effectiveness of the proposed method to reject torque disturbances, a simple model for the torque disturbance is added to the jMAVSim simulation model using

$$d_\tau = -c_\tau \cdot \omega \tag{35}$$

where $c_\tau = 0.3$ N ms = rad is the drag coefficient.

To evaluate the performance of the controller the reference trajectory is taken as figure-8 given by $p_d(t) = [A \sin(2\pi t/T) +$

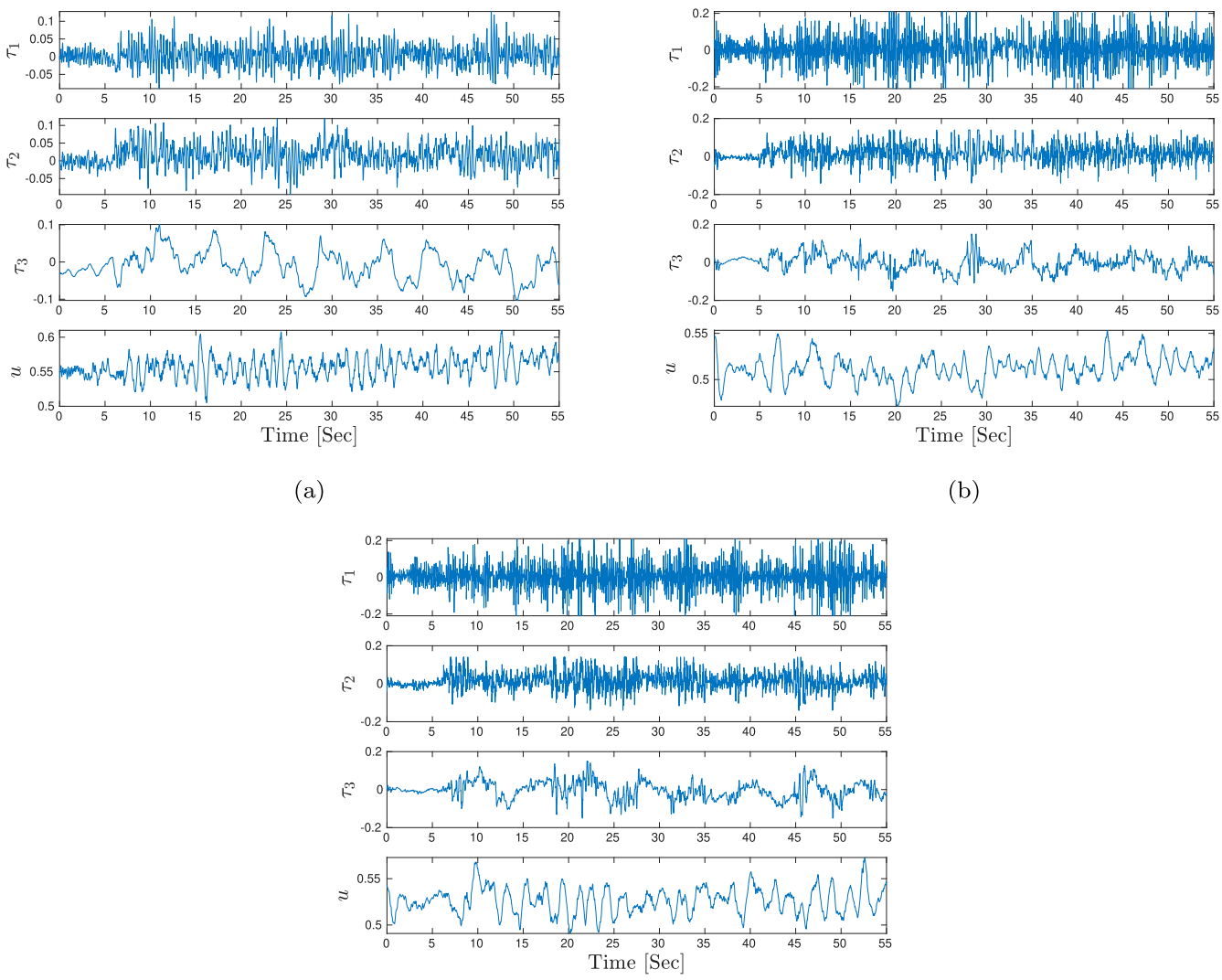

Figure 9. Experimental results: normalised control inputs. (a) PX4MC. (b) BS and (c) BSDOBC.

$1, B\sin(4\pi t/T), -0.85]^\top$ m, with $A = 1.5$ and $B = 0.75$. The velocity of the trajectory is increased by decreasing T from 20 at $t = 0$ to 12 in 8 seconds. The setpoint for yaw is zero. In order to investigate the benefit of the DO, the backstepping disturbance observer-based control (BSDOBC) is compared to the backstepping controller (BS) and the built-in PX4 position control (PX4MC). Extensive simulation was performed at different values for controller gains while monitoring the RMS value of the tracking error. This testing led to optimal values $k_1 = k_2 = k_3 = k_4 = 7$. We observed that lower values of gain generally led to larger RMS error while larger values destabilised the UAV. After tuning $k_1 = k_2 = k_3 = k_4$ a similar process was used to find optimal observer gains $k_{d_f} = 0.5$ and $k_{d_\tau} = 0.5$. The vehicle is initialised in hover at $p(0) = [0.5, 0.5, -1]^\top$ m. Hence, a relatively large initial error exists when the controller is switched on, i.e. $\tilde{p}(0) = [0.5, 0.5, -0.15]^\top$ m.

Simulation results for the BSDOBC are shown in Figure 4. As can be observed, the tracking error converges to an acceptable neighbourhood of the origin in approximately 10 s and remains there. The normalised torques and thrust inputs are in Figure 4(b). The inputs have reasonable magnitude and remain unsaturated throughout the simulation. As expected, the thrust has an average value close to 0.51 N. The disturbance estimates are shown in Figure 4(c,d). We observe $\hat{d}_{f_1}$ has a steady-state average value of approximately -0.1 N and contains oscillations at the frequency of the reference trajectory. The steady-state average value of $\hat{d}_{f_2}$ is about -4 N which is consistent with the applied wind disturbance in the $-n_2$ direction. The steady-state average value of $\hat{d}_{f_3}$ is almost 0.3 N which is due to modelling error. We observe the estimate for the torque disturbance is almost zero. This is consistent with the disturbance model (35) and the small angular velocity of the quadrotor along the figure-8 trajectory.

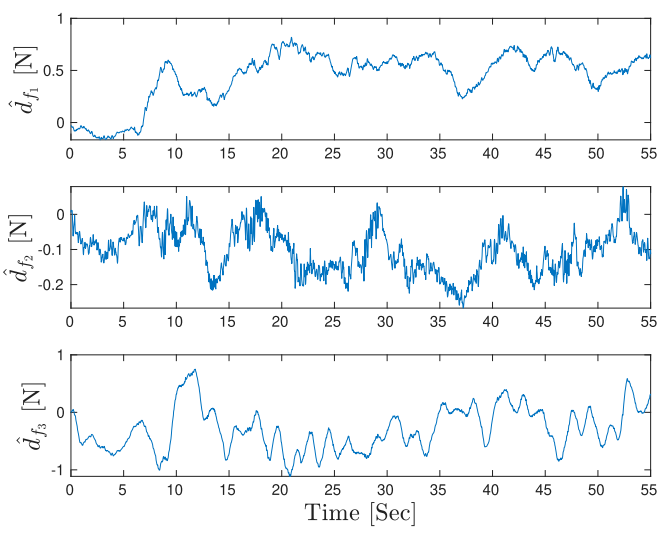

Figure 10. Experimental results: disturbance estimates.

The simulation results for the actual and desired trajectories are shown in 2D for all controllers in Figure 5. Table 2 provides the root mean square of the position error (RMSE) in steady-state. The trajectory of the norm of the tracking error is in Figure 6. We observe from Table 2 that the BSDOBC control achieves the lowest RMSE values. The 2D plots clearly reflect this reduced error and the effectiveness of the disturbance observer in improving the tracking performance. The code which reproduces this simulation is available online for public use (Moeini & Lynch, 2020).

4.4 Physical experiment

In this section, we investigate the performance of the proposed controller implemented on the ANCL quadrotor platform. External disturbances are created with a fan which was placed at a height of about 0.85 m (relative to the fan's centre) and 3 m away from the UAV's initial position. The fan has a blade diameter of about 0.6 m and it produces an airflow of about 1.6 m/s in the direction of the n_1 axis. We remark that the airflow due to the fan will alter the thrust generation of the vehicle, and this change in aerodynamics is difficult to model. As in the simulation, the reference trajectory is a figure-8 given by $p_d(t) = [A\sin(2\pi t/T) + 1, B\sin(4\pi t/T), -0.8]^\top$ m, with $A = 0.7$ and $B = 0.3$. The velocity of the trajectory is increased as a function of time by decreasing T from 20 to 12 in 8 seconds. The setpoint for yaw is zero. As in the simulation, the BSDOBC is compared with BS and PX4MC controllers.

The experiment consists of flying close to the starting point of the reference trajectory manually. Then the controller is switched to setpoint regulation mode (using the stock PX4 control) with setpoint $p_d(0) = [1, 0, -0.8]$ m. Next, one of the three trajectory tracking controllers is enabled and the fan turned on. The controller gains for the BSDOBC are in Table 3. In order to tune the controller gains, the lateral position subsystem was considered initially. We began with small values of k_1, k_2, k_3, k_4 and the remaining 2 DoF (i.e. yaw and altitude) controlled by the stock PX4 motion controller. The RMS value of the steady-state tracking error was observed as k_1, k_2, k_3, k_4 and torque normalisation parameters n_{τ_1}, n_{τ_2} and n_{τ_3} were systematically varied. A similar procedure was applied to tune the other 2 DoF which are affected by $k_{\psi 1}, k_{\psi 2}$ and the thrust normalisation (32).

The resulting tracking errors, p_1-p_2 plot, control inputs and disturbance estimates are shown in Figures 7–10, respectively. As can be observed from Figures 7 and 8, the BSDOBC provides better performance in the steady state compared with the BS and PX4MC controllers. We observe an offset about 20 cm in the n_1 direction for the tracking error of the BS controller. This offset is due to the applied force disturbance from the

Table 4. Experimental results: mean and RMSE of position tracking error.

Controller	$\tilde{p}_1$(mean)	$\tilde{p}_2$(mean)	$\tilde{p}_3$(mean)	$\tilde{p}_1$(RMSE)	$\tilde{p}_2$(RMSE)	$\tilde{p}_3$(RMSE)	$\|\tilde{p}\|$(RMSE)
PX4MC (m)	−0.0193	0.0037	−0.0337	0.2598	0.2036	0.0467	0.2810
BS (m)	0.1509	−0.0312	−0.0089	0.1796	0.0497	0.0475	0.1501
BSDOBC (m)	0.0433	−0.0101	−0.0010	0.0952	0.0380	0.0346	0.0668

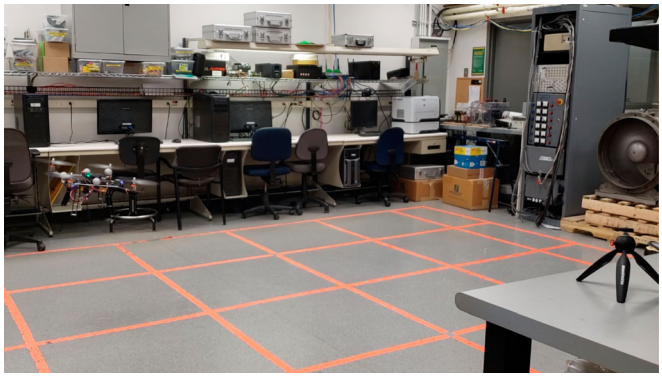

Figure 11. Experimental setup.

fan and since the BS control has no disturbance compensation. The force disturbance estimate is shown in Figure 10. The average steady-state value of $\hat{d}_{f_1}$ is about 0.5 N. From Figure 9, we observe the torques are noisy which is mainly due to the noise in angular velocity measurements. The steady-state average value and RMSE of the tracking error for the three methods are shown in Table 4. We observe the BSDOBC achieves better performance which is to be expected as it estimates and compensates for the disturbance force. Figure 11 shows a picture of the experimental setup. A video of the experiment is provided at https://youtu.be/-90w_FPmNWE.

5. Conclusion

This paper has proposed a nonlinear position and yaw tracking control which combines backstepping with a DO to increase its robustness to external force disturbances. The DO error dynamics is decoupled from the tracking error dynamics and the cascade structure of the closed-loop makes the stability analysis, implementation, and tuning straightforward. Exponential stability is proven for the constant disturbance case. The method is validated in SITL simulation and flight experiment using the commonly used PX4 autopilot firmware. The BSDOBC is compared with a stock PX4 inner-outer loop control and a traditional backstepping method. The BSDOBC demonstrates improved tracking performance and robustness relative to these controllers.

Disclosure statement

No potential conflict of interest was reported by the authors.

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